Poster: Towards Mitigating Unequal Membership-Inference Risk in GNNs with Self-Supervised Pretraining

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Abstract

Graph Neural Networks (GNNs) achieve strong performance across many graph-based tasks but remain vulnerable to membership inference (MI). Prior work has shown that graph structure plays a central role in shaping MI risk. Since nodes differ in their structural and feature properties, we ask whether they also face unequal MI risk and find that structurally isolated nodes and nodes with high intra-class feature dissimilarity are particularly exposed. Motivated by this uneven vulnerability, we make the case for systematically exploring self-supervised learning (SSL) as a privacy-mitigation strategy for graph machine learning. As a first step, we investigate contrastive SSL pretraining on a public graph before fine-tuning on private data. In standard GNN training, a low backbone learning rate disproportionately protects the most vulnerable nodes, while SSL preserves utility and provides additional privacy gains across node groups. We then integrate SSL into PrivGNN, improving the privacy–utility trade-off by reducing the privacy budget by more than 50% on benchmark datasets.

CCS Concepts

• **Security and privacy**; • **Computing methodologies** → **Neural networks**;

Keywords

Graph Neural Networks, Membership Inference, Self-supervised Learning

ACM Reference Format:

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1 Introduction

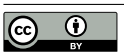
Many real-world graphs contain sensitive information, including patient relationships and financial transactions, and therefore cannot be shared directly. One way to share the knowledge gained from such data is to train and release a machine learning model rather than expose the underlying graph. However, machine learning models can memorize aspects of their training data, making them vulnerable to Membership Inference Attacks (MIAs), in which an adversary attempts to determine whether a particular individual was included in the training set [13].

Graph Neural Networks (GNNs) are particularly vulnerable to MIAs, and their vulnerability is shaped by the structure of the training graph [5, 10]. Prior work has studied how graph-level structural properties affect the overall success of membership inference against a GNN [5]. At the node level, He et al. show that membership inference vulnerability is unequally distributed across nodes [3]. We extend this work by drawing on the node-level properties introduced in [14], which characterize how such properties shape model behavior more broadly, and use them to explain this disparity. We find that the most vulnerable nodes are often structurally isolated or dissimilar from same-class peers, making their neighbourhood information less reliable.

For such nodes, learning useful representations directly from the private graph may require the model to adapt more strongly to their individual characteristics. This motivates us to investigate self-supervised learning (SSL) pretraining on a public graph [7]. Public-graph pretraining provides the model with a stronger representation prior before it is exposed to private data, potentially reducing its dependence on individual private nodes during fine-tuning. Unlike privacy mitigations such as random edge addition [3], which perturb the graph and may reduce predictive utility, our experimental results though limited indicate that SSL pretraining using unlabelled public graph followed by fine-tuning using private graph can improve both utility and empirical privacy robustness as compared to the models solely trained on the private graph. We specifically focus on the contrastive SSL method GRACE [15].

A key factor when fine-tuning an SSL-pretrained model is the backbone learning rate (BLR), which controls how quickly representations learned from the public graph are overwritten using private training data. We show that BLR acts as a tunable lever for navigating the utility–privacy trade-off. Lowering the BLR limits adaptation to private nodes and disproportionately protects the most vulnerable nodes. However, with random initialization, a low BLR also prevents the model from learning sufficiently from the downstream task. SSL makes a low BLR viable by providing a useful initialization that preserves predictive utility while reducing membership leakage.

We further study whether SSL can improve formal privacy guarantees when combined with Differential Privacy (DP) [1]. In particular, we integrate SSL into PrivGNN [9], a node-level DP framework based on knowledge distillation. PrivGNN trains a query-specific teacher model for each public query node. The teacher is trained on a induced subgraph over (K) private nodes most similar to that query node, and its noisy prediction is then used as a pseudo-label for training the public student model. Its privacy cost grows with



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the number of queries, while increasing the added noise severely reduces student accuracy. SSL provides the student model with representations learned from the public graph, making it less dependent on the noisy pseudo-labels produced by the private teachers. Consequently, SSL-integrated PrivGNN can reach a given level of utility using substantially fewer public query nodes. Because each query consumes privacy budget, this directly yields a tighter formal privacy guarantee without increasing the amount of noise or changing the deployed architecture.

2 Experimental Set-up

To provide preliminary evidence for our claims, we study standard GNN training and the differentially private PrivGNN framework under a common inductive setting. We first describe the construction of the private, public-training, and public-test graphs, together with the definitions of member and non-member nodes. We then present the threat model and target architecture, followed by the GRACE pretraining procedure, the backbone learning-rate design, and the integration of SSL into PrivGNN.

Datasets. We employ four benchmark datasets: Cora [8, 11], CiteSeer [11], PubMed [11], and Amazon [12]. For the PrivGNN experiments, we additionally consider ArXiv [4] and Reddit [2]. For each dataset construct three equally sized, disjoint graph partitions from the same underlying dataset: a labelled private graph $\mathcal{G}_{priv}$, an unlabelled public training graph $\mathcal{G}_{pub_train}$, and a public test graph $\mathcal{G}_{pub_test}$. Nodes are randomly assigned to the three partitions, and each graph is formed as the induced subgraph over its assigned nodes, removing all cross-partition edges.

For standard GNNs, we pretrain on $\mathcal{G}_{pub_train}$ and fine-tune only on $\mathcal{G}_{priv}$; members are private training nodes, while non-members come from the unseen $\mathcal{G}_{pub_test}$. For PrivGNN, unlike the original implementation, we train the student only on the induced graph over public training nodes.

Threat and Target Model. We use a black-box adversary with access only to model posteriors and ground-truth labels for 50% of the nodes. For each node, the attacker concatenates the predicted posterior with its cross-entropy loss and feeds this vector to a two-layer MLP to predict membership. Attack performance is reported using ROC AUC which in the following we refer to as MIA AUC (Membership Inference Attack AUC). We employ a 2-layer Graph Convolutional Network (GCN) [6] with a hidden dimension of 64, ReLU activations, batch normalization, and a dropout rate of 0.5. The linear classification head maps encoder representations to class logits. All models are trained for 200 epochs using Adam with a learning rate of 0.01.

2.1 Self-supervised Learning Set-up

Pretraining with SSL. We pretrain the GCN [6] encoder on the public graph $\mathcal{G}_{pub_train}$ using GRACE [15]. GRACE is a contrastive SSL technique that first generates two graph views using feature masking and edge dropping, then learns representations by pulling together the same node across views and pushing apart different nodes.

Backbone Learning Rate. The pretrained encoder is transferred to the downstream model and fine-tuned on private data together with a randomly initialized classification head. The backbone learning rate (BLR) controls how strongly private supervision updates the encoder: lower values preserve the SSL representations, while higher values overwrite them more aggressively. We fix the head learning rate at 10^{-2} and vary the encoder BLR.

Integrating SSL into PrivGNN. The pretrained encoder is transferred to the student model which (together with a randomly initialized classification head) is then fine-tuned using noisy weak supervision as in the original PrivGNN implementation. We fix the classification head learning rate at 10^{-2} and vary the encoder BLR.

3 Experiments and Results

We conduct three experiments. First, we study how the backbone learning rate and SSL jointly affect the privacy–utility trade-off. Second, we study how the attack performance varies with node properties. Third, we evaluate whether SSL enables PrivGNN to achieve comparable utility with fewer query nodes and, consequently, a lower privacy budget. We make the following observations.

Utility-privacy trade-off. In Figure 2 we vary the backbone learning rate (BLR) and compare randomly initialized and SSL-pretrained models in terms of MIA AUC and non-member accuracy for different datasets. The results suggest that SSL and BLR play complementary roles in the privacy–utility trade-off. At a low BLR of 10^{-4}, MIA AUC decreases mainly because confidence on member nodes is reduced, while non-member utility is only mildly affected. At a moderate BLR around 10^{-3}, SSL preserves this privacy benefit while improving non-member accuracy beyond the no-SSL baseline. As BLR approaches 10^{-2}, the pretrained representations are increasingly overwritten, causing both privacy and utility to converge towards the baseline.

Differing vulnerability of nodes and impact of SSL. In Figure 2 we study how membership inference risk varies with node degree. Our fine-grained analysis at the node levels reveals that a low BLR of 10^{-4} disproportionately protects the most exposed nodes. The largest reductions in attack success rate occur for low-degree, structurally isolated nodes, which face elevated MI risk [3], suggesting that low BLR limits overfitting to private node-specific signals. SSL then provides an additional, more uniform reduction in vulnerability across all node groups. We observe similar results (not presented due to space limitations) when using an intra-class feature dissimilarity metric from [14] instead of node degree, implying that nodes with atypical feature vectors for their class face higher MI inference risk and benefit most from low BLR.

Impact of SSL on PrivGNN. We evaluate PrivGNN at a fixed noise parameter of $\lambda = 0.2$, where λ is the inverse of the scale parameter of the Laplace distribution used to add noise when generating the noisy labels for the public train set. At this setting, SSL improves accuracy by 0.150, 0.139, and 0.291 on Reddit, ArXiv, and Amazon, respectively. As shown in Figure 3, SSL-augmented PrivGNN matches the baseline accuracy using only 294, 146, and 32 query nodes, corresponding to reductions of 71%, 85%, and 96%. This in turn reduces the privacy budget from 9.69/9.69/8.15 to 4.78/3.21/1.37 for the three datasets, without sacrificing utility.

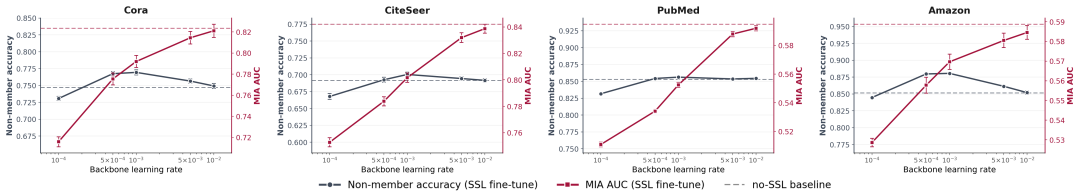


Figure 1: Utility-Privacy trade-off as a function of backbone learning rate (BLR) for SSL pre-trained model with GRACE.

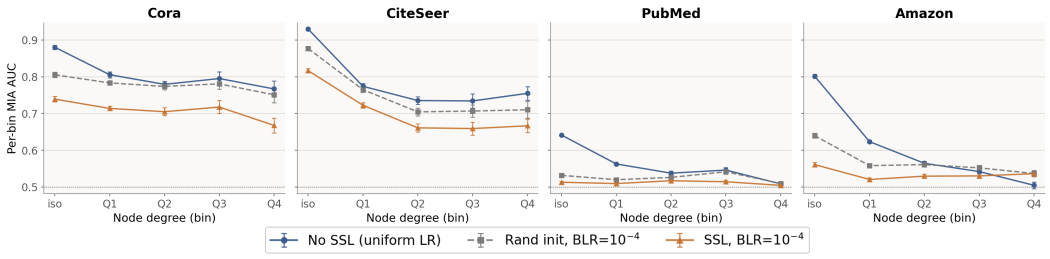


Figure 2: Impact of node degree on MIA AUC. Nodes are sorted by degree and divided into four equal-sized quartiles, ordered as $Q_1 < Q_2 < Q_3 < Q_4$, where Q_1 contains the lowest-degree nodes and Q_4 the highest-degree nodes. We compare three training regimes: no SSL with uniform learning rate (blue), no SSL with random initialised model training with backbone learning rate of 10^{-4} (gray) and pretraining with SSL with fine-tuning with backbone learning rate of 10^{-4} (orange)

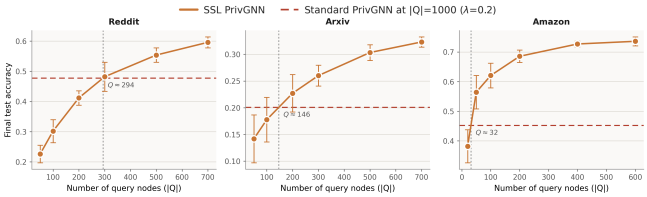


Figure 3: Non-member accuracy vs. $|Q|$ ($\lambda = 0.2$). Dashed : PrivGNN. vertical : point where SSL-PrivGNN matches the baseline.

4 Limitations and Future Work

As a first step towards a broader research agenda, we present preliminary evidence that public-graph SSL can improve the privacy–utility trade-off of GNNs. We acknowledge the limitations of this preliminary study: it uses equal-sized random partitions from the same graph to build train-test partitions and does not evaluate distribution shift, creating a favourable setting for SSL transfer in which privacy risk may be underestimated and utility overestimated. The analysis is also limited to a single SSL method and homophilic datasets. Our PrivGNN replication remains 5–15% below the originally reported utility, although the relative comparisons remain fair. Building on this, we plan to study controlled public–private distribution shifts, compare contrastive, generative, and predictive SSL methods, extend the analysis to heterophilic graphs, and integrate SSL into other differentially private frameworks.

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